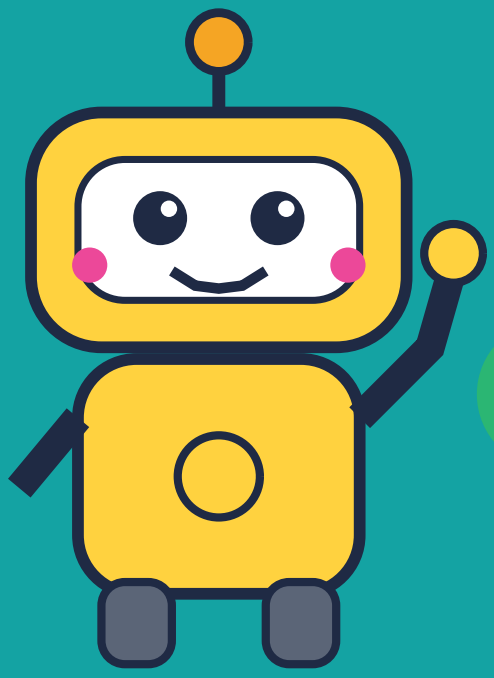
SPECIMEN BOOKLET

Think Like a Coder

Computational thinking and AI for Classes 6–8

A sample for schools: a CT chapter, an AI chapter, a full project, a teacher script, the classroom kit and how assessment works



For CBSE's Computational Thinking and AI curriculum, 2026-27

Class 6 sample · NCERT maths chapters, CBSE AI chapters and projects · No lab needed

Projects and AI Pack Rs 599 per student (indicative) · Free pilot, January-March 2027

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CodeSchool is a working name. Not an official CBSE or NCERT publication.

A complete CT&AI programme for Classes 6-8

From 2026-27, CBSE asks schools to teach Computational Thinking and AI in Classes 3-8. In Classes 6-8 it is **100 hours a year**; in Class 6 that is about 40 hours of computational thinking, 20 hours of AI and two projects of 20 hours each. CBSE's free handbook gives puzzles for each maths chapter and four AI chapters. CodeSchool turns it into a full course any teacher can run, with no computer lab needed.

What schools need	CBSE's free handbook	CodeSchool Projects and AI Pack
CT for each maths chapter	Puzzles	Theory, worked examples, practice in levels and coding on paper
AI chapters	Four chapters with exercises	The same four topics, with unplugged activities and 'build it' tasks
Coding	No	Functions, lists, loops and flowcharts on paper, ready for block or text coding
Projects (40 hours)	Hours set aside	Two complete 20-hour projects with plans, data sheets and rubrics
Teacher lesson plans	Answer keys	A period-by-period teacher script book
Assessment	No format given	Project rubrics, observation journal and term summary
Teacher training	CBSE workshops	School orientation, termly workshops and support

★ Why schools choose it

- ✓ Chapters 1-10 follow NCERT Class 6 maths in order, so the CT work fits the maths period.
- ✓ The four AI chapters match the AI part of CBSE's handbook and are taught with hands-on, unplugged activities.
- ✓ Two ready-made projects fill the 40 project hours, with plans, data sheets and rubrics.
- ✓ Works in any classroom: paper, cards and posters. A projector or computer lab helps but is optional.

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What each school receives

For	What they get	Classes
Each student	Projects and AI Pack: workbook with CT chapters, AI chapters, two project units and cut-out cards	6, 7, 8
Each class section	Classroom kit: AI picture cards, function and flowchart cards, data sheet pads, project posters, box	6, 7, 8
Each teacher	Teacher script book, answer keys, project rubrics, observation journal, short how-to videos	6, 7, 8
The school	Orientation at school, a workshop every term, a support line and a term summary format	All

The series

Book	Levels	Coder skills
Classes 3-5 Practice Packs	B1-B7	Steps, loops, if-else, variables, search and a game designed on paper (separate samples)
Class 6 Projects and AI Pack (this sample)	C1-C3	Functions, lists, sieve and sorting, debugging, four AI chapters and two projects
Class 7 Projects and AI Pack	C4-C5	In development, following CBSE's Class 7 handbook
Class 8 Projects and AI Pack	C6-C8	In development, following CBSE's Class 8 handbook

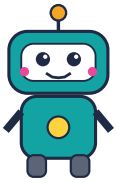
Price and ordering

Indicative price: Rs 599 per student per year (MRP) for the Classes 6-8 Projects and AI Pack, through the school's book list. It can replace an existing computer book, so the cost to parents stays about the same. Kits, teacher books and training come with the packs. Prices are indicative until confirmed in writing.

Try it first

A **free pilot** in two or three sections from January to March 2027: materials, a teacher orientation, weekly support and a short report against success measures agreed in advance. No obligation to continue.

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Hello again! I am Robo. This year we go further: we will write functions, work with lists and data, find out how AI learns from examples, and run two real projects. All you need is a pencil and a curious mind.

How this book works

Learn	Robo explains one big idea in simple words, with the new coder words to remember.
Example	A worked example shows the idea step by step.
Try it!	Five practice questions. Circle the answer or write it in the box.
Now you build it	Coding on paper: put code cards in order, fix bugs and write your own programs.
Challenge	A harder puzzle for when you are ready for more.
How did I do?	Colour a face to show how sure you feel.
Projects	Two 20-hour team projects: plan, collect real data, build, test, present and score your work with a rubric.

My code cards

Coders use cards like these. Each colour means something different. You will find cut-out cards at the back of the book.

Step: do this **Repeat: a loop** **If: a decision** **Define: a function**

When: an event

For teachers and parents

Chapters 1-10 go with the same chapters of the NCERT Class 6 maths textbook, so they can be used in maths periods. AI chapters 1-4 follow the AI part of CBSE's Class 6 handbook, and the two projects run across the year. CBSE sets 100 hours a year in Class 6: about 40 for computational thinking, 20 for AI and 40 for two projects. The coding here needs no computer: students write, trace and debug programs on paper, which prepares them for block and text coding. Answers are at the back. Encourage children to explain their thinking, not just the answer.

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How a lesson runs

Every lesson fits one 40-minute period and follows the same five steps, based on the PRIMM approach used in computing classrooms (predict, run, investigate, modify, make). Children learn the routine once and then focus on the thinking.

1. **Predict and run** 8 min
   Children predict what a set of steps will do, then act it out.

2. **Investigate** 8 min
   Pairs answer 'why' questions about how it worked.

3. **Modify** 8 min
   Children change the steps to do something new.

4. **Put in order** 8 min
   Jumbled code cards go back in the right order.

5. **Make** 8 min
   Pairs write their own program and swap with another pair.

How it supports CBSE's assessment

CBSE asks schools to assess CT&AI through written puzzles, group activities and a **Teacher Observation Journal**. Each step gives the teacher something to observe: predictions show reasoning, modifications show understanding, and the 'make' step shows whether a child can build an algorithm of their own. The journal format and a term summary for the report card come with the teacher book.

Every child can take part

- ✓ Children who finish early move to the challenge level; children who need help use a hint card.
- ✓ Most activities are done in pairs, so children who find reading hard can still take part by talking and acting out.
- ✓ Teacher preparation is about 5-10 minutes per lesson using the script.

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Inside the full Class 6 book

The complete book has 10 CT chapters that follow the NCERT Class 6 maths textbook, 4 AI chapters that follow CBSE's handbook and two 20-hour projects, plus answers, a glossary and cut-out code cards. This sample includes CT chapter 1, AI chapter 3 and Project 1 in full.

Ch	Chapter	Coder skill	Level	In this sample
1	Patterns in Mathematics	Rules and functions: name a rule and reuse it	C1	Yes
2	Lines and Angles	Drawing programs: angles, turns and a square(size) function	C1	
3	Number Play	Algorithms that repeat until something happens	C1	
4	Data Handling and Presentation	Lists: store, sort and find the biggest	C1	
5	Prime Time	The sieve and a factor-finding algorithm	C2	
6	Perimeter and Area	Functions with two inputs; breaking shapes apart	C2	
7	Fractions	Compare, simplify and sort fractions	C2	
8	Playing with Constructions	Construction scripts: exact steps and debugging	C2	
9	Symmetry	Reflect and rotate: functions that change shapes	C2	
10	The Other Side of Zero	Integers in data: lowest, highest and range	C2	
AI1	Introduction to AI	Tell AI from automation; how machines learn	C3	
AI2	Basic Data Concepts	Types of data; collect, organise and show it	C3	
AI3	Patterns and Decisions	Learn a rule from labelled examples, then decide	C3	Yes
AI4	Ethics and Digital Responsibility	Strong passwords, privacy and digital footprints	C3	
P1	Weather Across India	Collect, organise and present real data	C3	Yes
P2	A Smarter School Routine	Design, test and improve an algorithm	C3	

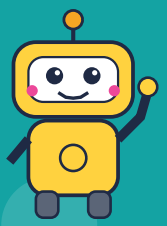
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CHAPTER 1 · LEVEL C1

1 Patterns in Mathematics

Coder skill: Rules and functions: name a rule and reuse it

Goes with NCERT maths, Class 6, chapter 1

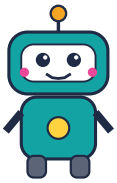


Learn

A **sequence** is a list of numbers that follows a **rule**. Some rules say how to get the **next** term (add 3 each time). Others say how to get the term at any **position** (term = 3 x position + 2), which lets you jump straight to the 100th term.

Coders give a useful rule a name and call it a **function**. A function takes an **input** and gives back an **output**: *square(4)* gives back 16, *triangle(4)* gives back 1 + 2 + 3 + 4 = 10.

sequence numbers that follow a rule	**term** one number in a sequence	**function** a named rule with an input	**output** what the function gives back



A rule I use again and again gets a name: a function. square(n) means n x n. Give it 5 and it gives back 25!

Example

Adding odd numbers makes squares

1. 1 = 1, 1 + 3 = 4, 1 + 3 + 5 = 9, 1 + 3 + 5 + 7 = 16. The answers are **square numbers**.
2. The picture shows why: each new odd number is an L-shaped layer that turns a square into the next square.
3. So the first **n** odd numbers add up to **n x n**. The first 10 odd numbers add up to 100.



★ *Coder fact!* Spreadsheet formulas work like functions: write the rule once and it fills thousands of rows.

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Try it!

Circle the right answer, or write it in the box.

1 Write the next two terms: 1, 4, 9, 16, 25, ...

2 Triangular numbers: 1, 3, 6, 10, 15, ... What comes next?

3 A sequence starts at 5 and adds 4 each time. What is its 10th term?

a) 41 **b)** 45 **c)** 40 **d)** 37

4 The function **f(n) = 3 x n - 1**. What is f(4)? Which input n gives the output 29?

5 In 2, 3, 3, 4, 4, 4, 5, 5, 5, 5, ... each number appears one time fewer than its value. How many times does 7 appear?

a) 5 **b)** 6 **c)** 7 **d)** 8

Now you build it

Write a function called **triangle(n)** that adds 1 + 2 + ... + n and gives back the total.

Define triangle(n) **Set total to 0** **Repeat for k = 1 to n** **change total by k** **give back total**

Run triangle(4) and triangle(6) by hand. What does each give back?

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★ Challenge

Find the sum of the first 20 odd numbers without adding them one by one. Explain your shortcut.

✓ How did I do?

I can describe a sequence with a rule.	I can use a function with an input.	I can spot square and triangular numbers.
		

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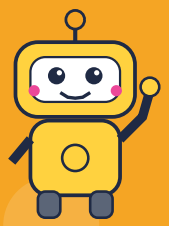
AI3

AI CHAPTER 3 · LEVEL C3

Patterns and Decisions

AI skill: Learn a rule from labelled examples, then decide

Goes with the AI part of CBSE's Class 6 CT&AI handbook, chapter 3

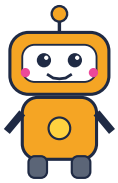


Learn

A **pattern** is something that repeats or can be predicted. AI finds patterns in **labelled examples** and uses them to **predict** labels for new things. This is how a spam filter learns which emails are spam.

An **observation** says what the data shows (it was warmer at 2 pm on 5 days). A **conclusion** says what it means (afternoons are usually warmer). A conclusion from a little data may be wrong, so we test it, and use more and fairer data.

pattern	observation	conclusion	prediction
something that repeats	what the data shows	what the data means	a best guess for a new case



I learn like this: look at examples with the right answers, find the pattern, make a rule, then test it on new examples!

Example

Learning when a mango is ripe

1. Look at the labelled examples. Colour does not decide it: mango 4 is yellow but not ripe, and mango 5 is green but ripe.
2. Every ripe mango is soft and every unripe one is hard. Rule: **if soft, then ripe**.
3. Test it on new mangoes whose labels we know. If it is right on most of them, we can use it to decide.

Mango	Colour and feel	Label
1	yellow, soft	ripe
2	green, hard	not ripe
3	yellow, soft	ripe
4	yellow, hard	not ripe
5	green, soft	ripe

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Try it!

Circle the right answer, or write it in the box.

1 'On 5 days it was warmer at 2 pm than at 8 am.' This is ...

a) an observation **b)** a conclusion

2 'So afternoons are usually warmer than mornings.' This is ...

a) an observation **b)** a conclusion

3 Labelled emails: *Win a prize now!* spam; *Class test on Monday* not spam; *Free gift, click here* spam; *PTM at 10 am* not spam. What would a filter trained on these predict for *Click now to win free coins*?

a) spam **b)** not spam

4 An AI learnt 'cat' only from photos of white cats. It is shown a black cat. What might go wrong?

a) It may not recognise it as a cat **b)** It will always be right

c) It will turn the photo white

5 Riya scored 18, 19 and 20 in weeks when she practised daily, and 12 and 11 in weeks when she did not. What is the best decision?

a) practise daily **b)** stop practising

c) nothing can be decided

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Now you build it

Be the classifier. Find the rule that tells a **Zib** from a **Zob**.

Training examples (labelled)



New creatures to classify



Look at the new creature | If it has ___ | predict Zib | else | predict Zob

Fill in the rule, then label the three new creatures. Does colour matter?

Challenge

A new creature has both stripes and spots. What should the classifier say? How could we make it better?

How did I do?

I can tell an observation from a conclusion.	I can find a rule from labelled examples.	I can explain why more and fairer data helps.
☺ 😐 ☹	☺ 😐 ☹	☺ 😐 ☹

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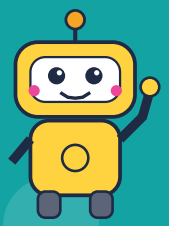
P1

PROJECT 1 · 20 HOURS · LEVEL C3

Weather Across India

Project skill: Collect, organise and present real data

CBSE: 40 project hours a year in Class 6, as two projects of 20 hours



The big question

How different is the weather in five Indian cities, and what patterns can we find?

Your team picks five cities in different parts of India, for example Leh, Shimla, Jaipur, Kolkata and Chennai. For two weeks, record each city's highest and lowest temperature and whether it rained, from the newspaper or an official source such as the India Meteorological Department (IMD).

Then organise the data, find the highest, lowest and range for each city, draw charts, look for patterns and present what you found. This project uses lists, integers, min and max, charts, patterns and conclusions.

dataset	source	range	trend
all the data you collected	where the data came from	highest minus lowest	a change over time

✓ My project plan

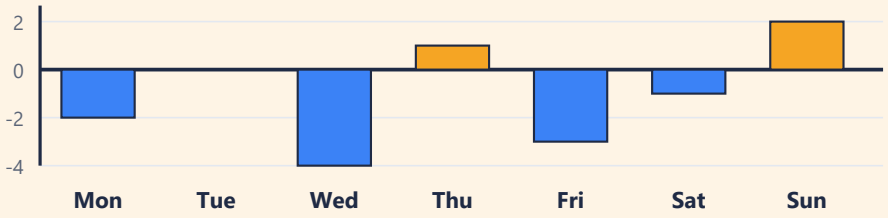
Weeks	Stage	What I do	Done
1	**Define**	Agree the question, choose five cities and decide what to record each day.	☐
2-3	**Collect**	Record the highest and lowest temperature and rain for each city every day for two weeks. Write the source.	☐
4	**Organise**	Put the data in one table per city. Look for missing or odd values and check them.	☐
5	**Calculate**	For each city find the highest, the lowest and the range. Count the rainy days.	☐
6	**Show**	Draw a bar chart of the ranges, and a bar chart of one city's daily highs.	☐
7	**Find patterns**	Write three observations and one conclusion. Is two weeks of data enough to be sure?	☐
8	**Predict and test**	Predict tomorrow's highest temperature for one city, then check the real value.	☐
9	**Present**	Make a poster or a 3-minute talk: question, data, charts, findings and limits.	☐
10	**Reflect**	Score your work with the rubric and write what you would improve.	☐

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How a coder does it

From raw data to a finding: Leh, one week in January (practice data)

1. Daily highest temperatures: -2, 0, -4, 1, -3, -1, 2 °C.
2. Highest 2, lowest -4, so the range is 2 - (-4) = **6 degrees**.
3. Observation: the highest temperature was below 0 °C on 4 of the 7 days.
4. Conclusion, with its limit: Leh's January days are often below freezing, but one week of data is not enough to be sure.



My data sheet

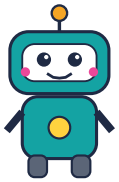
One row per day for one city. Write where each value came from.

Date	City	High °C	Low °C	Rain? (Y/N)	Source

Draw your chart

Draw a bar chart of the temperature range of each city. Choose a scale, then label both axes and give the chart a title.

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A project is like a big program: plan it, break it into parts, test it and fix it. One stage at a time!

★ How good is my project?

Score each row from 1 (just starting) to 4 (excellent). Then your teacher scores it too. Talk about any row where the scores differ.

What we check	What excellent (4) looks like	Me	Teacher
Question	Clear, and can be answered with the data collected.	1 2 3 4	1 2 3 4
Data	Complete, from a named source, with odd values checked.	1 2 3 4	1 2 3 4
Thinking	Highest, lowest and range are right; patterns are spotted.	1 2 3 4	1 2 3 4
Charts and talk	Charts have titles, scales and labels; the talk is clear.	1 2 3 4	1 2 3 4
Honesty and limits	Says what the data cannot show; nothing copied without credit.	1 2 3 4	1 2 3 4
Teamwork	Everyone has a job and the work is shared fairly.	1 2 3 4	1 2 3 4

✓ Fair, safe and honest?

- ☐ I wrote down where every piece of data came from.
- ☐ I did not change any data to make a better story.
- ☐ I said what my data cannot tell us (only two weeks, only five cities).
- ☐ I used only official or trusted websites, with an adult's help.

Looking back

What surprised you most in your data?

If you did this project again, what would you change?

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Sample teacher script: Train the classifier

From the teacher book, Class 6, AI chapter 3 (*Patterns and Decisions*). One 40-minute period. Level C3.

You need	
Kit	AI picture cards: 12 Zib and Zob creature cards (8 labelled for training, 4 unlabelled for testing), scoring slips
Book	AI chapter 3 pages, 'Try it' questions 3 and 4
Goal	Students find a rule from labelled examples, test it on new examples, count how many it gets right, and explain why varied data helps

Step and time	Teacher does	Children do
1. Predict and run (8 min)	Show the 8 labelled training cards. Ask: 'What makes a Zib a Zib?'	Groups list features (colour, stripes, spots) and predict a rule.
2. Investigate (8 min)	Ask: 'Does colour matter? How do you know?'	Groups check every card against their rule and drop features that do not fit.
3. Modify (8 min)	Hand out the 4 unlabelled test cards.	Groups classify them with their rule; the teacher reveals the labels and groups count how many were right.
4. Put in order (8 min)	Give out jumbled step cards: collect examples, find the pattern, make a rule, test, improve.	Order the steps of training an AI model and match them to what they just did.
5. Make (8 min)	Show a creature with both stripes and spots.	Decide what the classifier should say, and write one way to improve the training data.

Lesson set-up



Set-up: lay the labelled cards out as the 'training' set. Colours are mixed on purpose, so the only real pattern is stripes (Zib) or spots (Zob).

Common mistakes to watch for

- ✓ **A rule from one card.** 'Orange ones are Zibs' fits one card but not the others. Check the rule against every card.
- ✓ **Using the test cards to make the rule.** Then the score is not a fair test of what was learnt.
- ✓ **Thinking AI is always right.** A rule learnt from a few examples can fail on something new, like a creature with both patterns.

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Different paces

Support: start with 4 training cards that differ only in pattern.
Stretch: add a creature with a new feature (horns) and ask whether the rule must change; write the score as a fraction, such as 3/4 right.

Check for understanding

Two quick questions at the end of the period:

1. Why do we test on cards the classifier has not seen? (to check it learnt the pattern, not the answers) 2. A classifier got 9 of 12 test cards right. What fraction is that in simplest form? (3/4)

Note in the observation journal: **Beginning** (sorts by one card), **Developing** (finds a rule with help), **Secure** (finds and tests a rule and counts the score), **Extending** (explains the limits of the data and suggests better examples).

Curriculum link

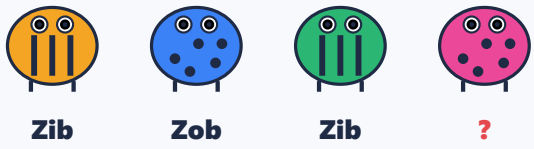
CBSE's Class 6 handbook introduces labelled data, supervised learning, observations and conclusions in its AI chapters. In this lesson students do by hand what a supervised learning model does, before they meet AI tools.

Teacher notes

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The classroom kit

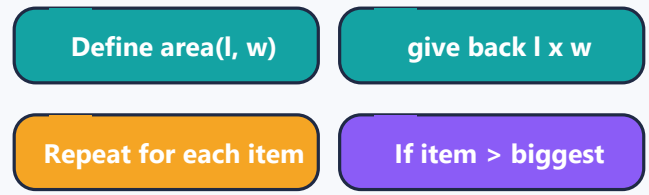
One kit per class section, used all year. Everything is printed, laminated and reusable.



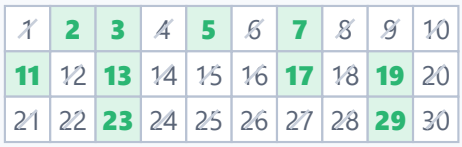
AI picture cards for classifying and testing (60)



Flowchart stencil cards for projects (30)



Function and loop cards (4 decks)



Sieve and data boards, write-on, wipe-off (10)

Item	Quantity per kit
AI picture cards (classify, train and test)	60
Function, loop and decision card decks	4 decks of 54
Flowchart stencil cards	30
Project data sheet pads (A4)	2 pads of 50
Write-on sieve and data boards	10
Project posters: plan, rubric, safe online	3
Teacher quick-guide card	1
Sturdy box	1

Illustrations show the design. Final materials will be confirmed after the pilot.

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Assessment and the report card

CBSE asks for continuous, competency-based assessment in CT&AI. The teacher book includes an observation journal using four levels, and a term summary that fits on the report card.

Skill	Beginning	Developing	Secure	Extending
Algorithms and functions	Follows given steps	Writes steps with loops	Writes and reuses a function	Designs and tests functions with inputs
Data	Reads a table	Organises data in a table or list	Charts data; finds lowest, highest, range	Draws fair conclusions and states limits
AI ideas	Names AI examples	Tells AI from automation	Explains learning from labelled data	Explains bias and how better data helps
Projects	Takes part with help	Completes stages with help	Plans, tests and presents with the team	Improves the work using evidence

Sample term summary

Area	Level	Teacher comment
Computational thinking (overall)	Secure	Writes functions with inputs and traces lists confidently.
AI and digital responsibility	Developing	Tells AI from automation; next step: explaining why data can be biased.
Project 1: Weather Across India	Rubric 19 of 24	Careful data collection with sources; charts need clearer scales.

Children check themselves too

Each chapter ends with **'How did I do?'**, where children colour a face for each 'I can' statement. Teachers can compare this with what they observed, which helps children learn to judge their own progress.

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Answers for the sample chapters

1. Patterns in Mathematics

1. 36 and 49
2. 21 (add 6)
3. a) 41 (5 + 9 x 4)
4. f(4) = 11; n = 10
5. b) 6
Build it: triangle(4) = 10; triangle(6) = 21
Challenge: 400: the first n odd numbers add up to n x n, and 20 x 20 = 400

AI 3. Patterns and Decisions

1. a) an observation
2. b) a conclusion
3. a) spam (it has the words 'win', 'free' and 'click')
4. a) it may not recognise it: its training data was not varied enough
5. a) practise daily
Build it: If it has stripes, predict Zib, else Zob. New creatures: Zib, Zob, Zib. Colour does not matter: each colour appears in both groups
Challenge: It cannot be sure, because it never saw one like this; it should say 'not sure', and we should add labelled examples like it to the training data

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Cut-out code cards

Ask an adult to help you cut along the dotted lines. Use the cards to build programs on your desk, then act them out with a friend.

Define ___	give back ___	Repeat ___ times
Define ___	give back ___	Repeat ___ times
Repeat for each item	Repeat until ___	If ___
Repeat for each item	Repeat until ___	If ___
else	When ___ happens	Input ___
else	When ___ happens	Input ___
Set ___ to ___	change ___ by ___	say ___
Set ___ to ___	change ___ by ___	say ___

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A note for parents

Schools can send this page home at the start of the year.

Dear parents,

This year your child will learn **computational thinking and AI**, following CBSE's new curriculum for Classes 3-8. In Class 6 that is about 100 hours over the year: problem solving linked to maths, four short AI chapters and two group projects.

Your child will use the *Think Like a Coder* workbook. Most of the work is on paper: writing exact step-by-step instructions, working with real data, learning how AI finds patterns, and learning to use technology safely and honestly.

How you can help at home

- ✓ Ask your child to explain a project to you: the question, the data and what they found.
- ✓ Talk about AI you use at home, such as voice assistants and video suggestions. What data might they use?
- ✓ Agree family rules for passwords, sharing photos and talking to strangers online.
- ✓ Praise careful thinking and honest results, even when a project does not go to plan.

Take-home challenge

Write down the highest temperature in your town every day for a week, from the newspaper or the weather report. What were the highest, the lowest and the range?

Answer: answers vary; the range is the highest minus the lowest.

No child's personal data is collected by the workbook. Thank you for your support.

Class teacher _____ **School** _____

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Next steps for your school

1. **Look through this sample**
   Try a chapter with a class or ask a teacher to review it.

2. **Talk to us**
   A short meeting with the principal and coordinator to answer questions.

3. **Start a free pilot**
   Two or three sections, January to March 2027, with an orientation and weekly support.

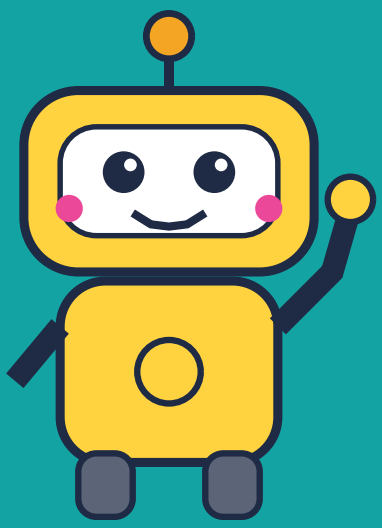
4. **Add it to the 2027-28 book list**
   Most schools finalise their list by the end of January.

Contact

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Notes

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Think Like a Coder

Computational thinking and AI for Classes 6-8, for CBSE schools

Ask us about a free pilot

Phone / WhatsApp: [Phone] · Email: [Email]

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